MA321 Quiz 3 Review Answers

1. % Error = Abs(Difference) ÷ Actual.

Month	Actual	Forecast	Difference	Abs(Difference)	% Error
1	50	45	5	5	10%
2	80	88	−8	8	10%
3	40	50	−10	10	25%

MAPE = (10% + 10% + 25%) ÷ 3 = **15%**

2. Answer True or False for each.

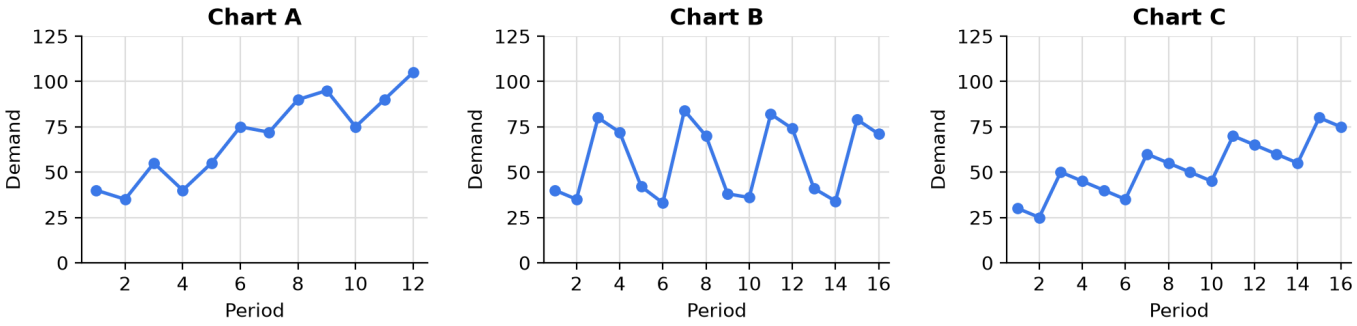


Chart A: upward trend, not seasonal, lots of noise.
Chart B: no trend (stationary), seasonal (repeats every 4 periods), little noise.
Chart C: upward trend, seasonal (repeats every 4 periods), little noise.

a) **True.** Chart B has no trend; it moves up and down around the same level.

b) **False.** Chart A has no repeating pattern; its ups and downs are noise.

c) **True.** Chart C goes up over time.

d) **True.** Chart A jumps around a lot; Chart C follows a clean repeating pattern.

e) **False.** Chart C has an upward trend, so it is not stationary.

3. Forecast = Baseline × Seasonal Index. For b, the baseline is 20% lower: $20 \times 0.8 = 16$.

Month	Seasonal Index	a) Forecast (baseline 20)	b) Forecast (baseline 16)
1	0.6	$20 \times 0.6 = \mathbf{12}$	$16 \times 0.6 = \mathbf{9.6}$
2	1.2	$20 \times 1.2 = \mathbf{24}$	$16 \times 1.2 = \mathbf{19.2}$
3	0.95	$20 \times 0.95 = \mathbf{19}$	$16 \times 0.95 = \mathbf{15.2}$
4	1.1	$20 \times 1.1 = \mathbf{22}$	$16 \times 1.1 = \mathbf{17.6}$
5	1.4	$20 \times 1.4 = \mathbf{28}$	$16 \times 1.4 = \mathbf{22.4}$
6	0.75	$20 \times 0.75 = \mathbf{15}$	$16 \times 0.75 = \mathbf{12}$

Each forecast in b is 20% lower than the one in a.

4.

a) **Forecast 2.** The lower MAPE (12%) is more accurate.

b) The forecast graph stays very close to the actuals graph. On average the forecast is only 3% off from the actual demand.

c) **True.** If the forecasts are far off (for example, forecasting 30 when the actual is 10 is a 200% error), the MAPE can be over 100%.

5. Deseasonalized demand $= \dfrac{\text{Actual}}{\text{Index}}$

Month	**Actual**	**Index**	**Deseasonalized**
January	36	0.8	$\dfrac{36}{0.8} = \mathbf{45}$
February	54	0.9	$\dfrac{54}{0.9} = \mathbf{60}$
March	50	1.0	$\dfrac{50}{1.0} = \mathbf{50}$
April	60	1.2	$\dfrac{60}{1.2} = \mathbf{50}$
May	63	1.5	$\dfrac{63}{1.5} = \mathbf{42}$
June	30	0.6	$\dfrac{30}{0.6} = \mathbf{50}$

b) **Best: February** (60), the highest deseasonalized demand. **Worst: May** (42), the lowest.

May had the highest actual sales (63), but May is expected to be a big month (index 1.5), so it actually underperformed. June had the lowest actual sales (30), but June is a slow month (index 0.6), so it performed at an average level (50).